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Right-F280 -- Update for F280-12

Wrightsoft has added F280-12 calculation procedures to Right-Suite Universal (RSU), starting at version 15.0.14.

This implementation is based on *Determining the required capacity of residential space heating and cooling appliances,* Standard F280-12, CSA Group, Mississauga, Ontario, Canada, March 2014. Additional interpretation information was derived from *Residential Heat Loss and Heat Gain Calculations, 2014 Edition,* Heating, Refrigeration, and Air Conditioning Institute of Canada (HRAI), 2014.

We have built into RSU all data and procedures required for performing F280-12 calculations, including weather data, the various load calculation factors found in the CSA standard, and supporting calculations. In particular, the CSA procedure makes use of external Excel spreadsheets AIM2.xls, SlabOnGradeHLR.xls, BasementHLR.xls (for calculating, respectively, infiltration air leakage, slab-on-grade heat loss, and basement heat loss). These methods are included in RSU, so there is no need to perform side calculations and transfer results.

Please note the following when using RSU for F280-12 calculations. In some cases, RSU input procedures are a bit different than those found in HRAI or the CSA Standard.

Load Calculation Method. To enable F280-12 calculations, select "CSA / CAN F280-12 (Right F280)" under Options | Load calculation method. You can set the default method via "Change default method" on the same menu.

Unit System. RSU operates in either SI (Metric) or I-P (Imperial) units. The F280-12 Standard is written in SI; we have applied conversions as needed. Select the desired system on the Options menu. This selection can be changed at any time. The shortcut Ctrl-U toggles the unit system between the systems.

Above-grade portions of basement walls. The spreadsheet BasementHLR.xls calculates basement heat loss for both the above- and below-grade wall areas. In RSU, the above-grade portions should be entered as exterior walls of basement rooms (with windows as appropriate). The below grade portion of the walls are generated when the basement floor configuration is selected. The area of the above grade portions is automatically derived from the room height and the basement depth below grade.

Headers. HRAI worksheets provide for entry of header (rim joist) areas. Both HRAI and the F280-12 Standard provides guidance on determining the effective R-value of header constructions. If you want to explicitly consider headers in your load calculation, you may include them as separate wall areas having suitably R-value. In well insulated buildings, the separate consideration of headers is only a minor effect relative to overall wall loss, so a common approximation is to use the floor-to-floor height as the room height and thus include headers in wall areas.

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Window properties. In this initial release, the RSU radio button system for describing windows does not offer all configurations documented in the F280-12 Standard and HRAI. The Custom Value screen can be used to enter any window characteristics. More radio button choices will be added in a future version.

Solar Corrections. When calculating opaque surface heat gain, RSU includes effective temperature corrections to account for daily temperature swings and solar heating of exposed surfaces. The corrections are given in Table 3 of the F280-12 standard. HRAI duplicates these values in their Table 11. However, Table 11 is described as the "detailed procedure" and a simpler "normal procedure" set of corrections is provided in Table 10. Since the HRAI Table 10 procedure is not found in the F280-12 Standard, it has not been implemented in RSU. Also note that since some of the corrections are negative, the calculation procedure can yield negative cooling loads in locations with moderate cooling design temperatures.

Internal gains. For cooling load calculations, F280-12 accounts for heat gain from two sources –

- Occupants at 70 W per person that is expected to be present during peak load times. These are assigned to the rooms where they typically are (generally living areas).
- Lights, appliances, and electrical plug-load heat gains are calculated on the basis of floor area and assigned to rooms that are designated "electrical peak load rooms."

The room assignments for heat gains are not done automatically. When preparing input, make sure you locate occupants appropriately and indicate that rooms such as kitchens and family rooms are flagged as "electrical peak load."

Ventilation. Load calculations must account for the effects of mechanical ventilation (supply, exhaust, or both). The required quantity of ventilation fresh air must be determined according to requirements of the local jurisdiction. This amount should be the principal or continuous rate; flows due to kitchen hoods or other intermittently-operated fans should generally not be included in load calculations. If the house has a separate ventilation duct system, then design air flow rates should be entered on each room. Otherwise, the central system air flow rate is entered via the Zone Information screen. Heat Recovery Ventilator (HRV) effectiveness can be entered for either configuration via Zone Information.

Report. The mandatory report "Heat loss and gain calculation summary sheet" can be generated by selecting Loads: F280-12 H/C Summary on the Report Builder tab. Typically this would be the first report included in a permit submittal package. Other standard RSU reports should be used to provide supporting information as required by your jurisdiction.